

VAR-X Model Report

Generated 2026-07-15 02:43

Model equation:

$$y_t = c + \sum_{i=1}^p A_i y_{t-i} + B x_t + u_t$$

$y_t \in \mathbb{R}^2$ (endogenous), $x_t \in \mathbb{R}^2$ (exogenous), $p = 1$

Endogenous: y1, y2

Exogenous: x1, x2

Differencing applied to endog: 1x

Observations: 179 AIC: 1035.76 BIC: 1077.20 Converged: True

Estimated coefficients ($L\{\text{lag}\}.\{\text{predictor}\}.\{\text{response}\} / \text{beta}.\{\text{exog}\}.\{\text{response}\}$)

	coef	std_err	pvalue
intercept.y1	-0.0834	0.0797	0.2954
intercept.y2	-0.1012	0.0795	0.2029
L1.y1.y1	-0.0196	0.0737	0.7907
L1.y2.y1	-0.0791	0.0765	0.3016
L1.y1.y2	-0.0167	0.0781	0.8309
L1.y2.y2	0.0487	0.0751	0.517
beta.x1.y1	-0.071	0.0831	0.3932
beta.x2.y1	0.1294	0.0945	0.1711
beta.x1.y2	-0.0185	0.0839	0.8256
beta.x2.y2	-0.0391	0.0878	0.6562
sqrt.var.y1	1.0065	0.0563	0.0
sqrt.cov.y1.y2	0.0405	0.0793	0.6092
sqrt.var.y2	0.9764	0.0594	0.0

ADF tests - levels

	adf_stat	pvalue	n_lags	stationary
y1	-1.4171	0.574	0	False
y2	-1.8966	0.3336	0	False

ADF tests - transformed (as modelled)

	adf_stat	pvalue	n_lags	stationary
y1	-14.3231	0.0	0	True
y2	-13.1805	0.0	0	True

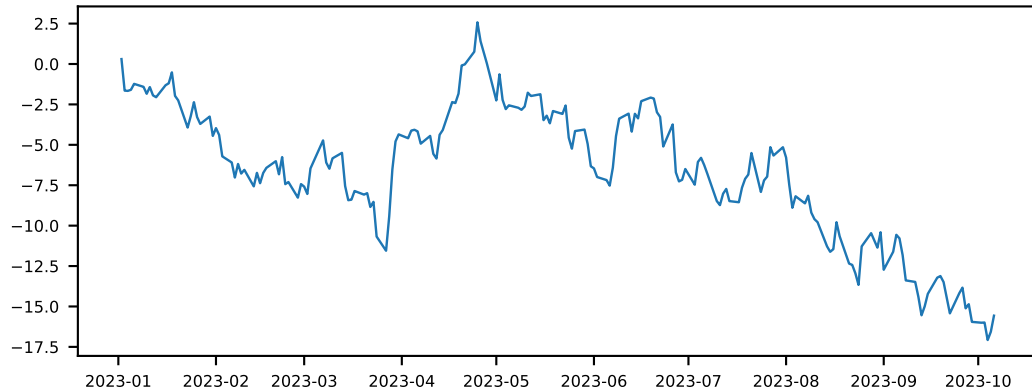
Lag order selection (information criteria)

	aic	bic	hqic	converged
1	1035.7607	1077.1968	1052.5627	True
2	1042.8002	1096.9858	1064.7721	True
3	1042.5585	1109.4936	1069.7002	True

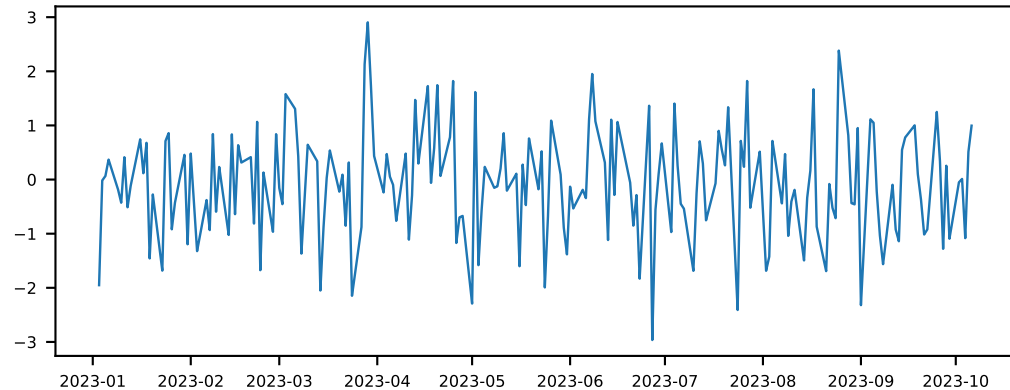
Residual diagnostics

	durbin_watson	lb_pvalue	dw_verdict
y1	1.982	0.6784	ok
y2	1.994	0.836	ok

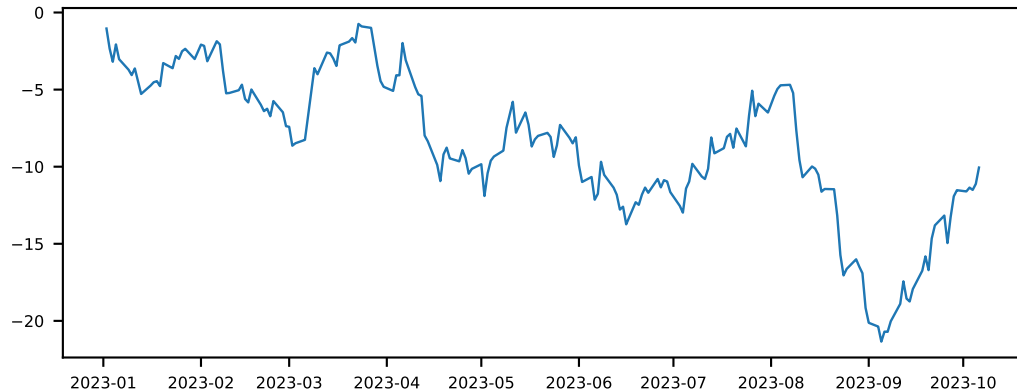
y1 (level) - ADF p=0.5740 [NON-stationary]



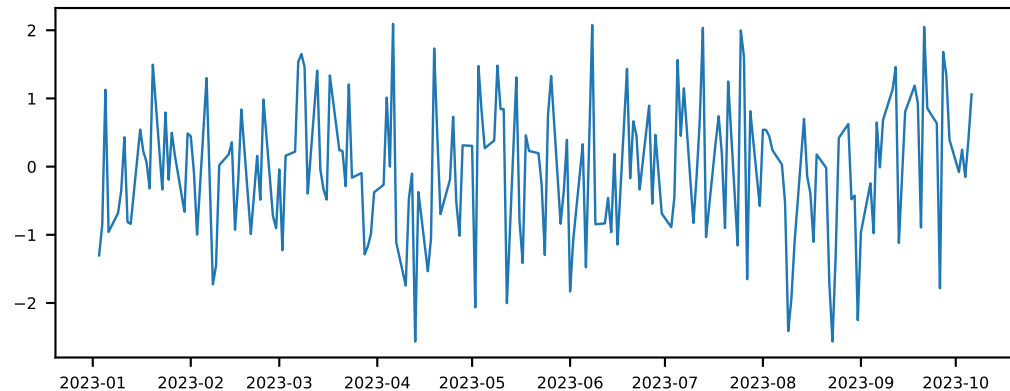
y1 (1st difference) - ADF p=0.0000 [stationary]



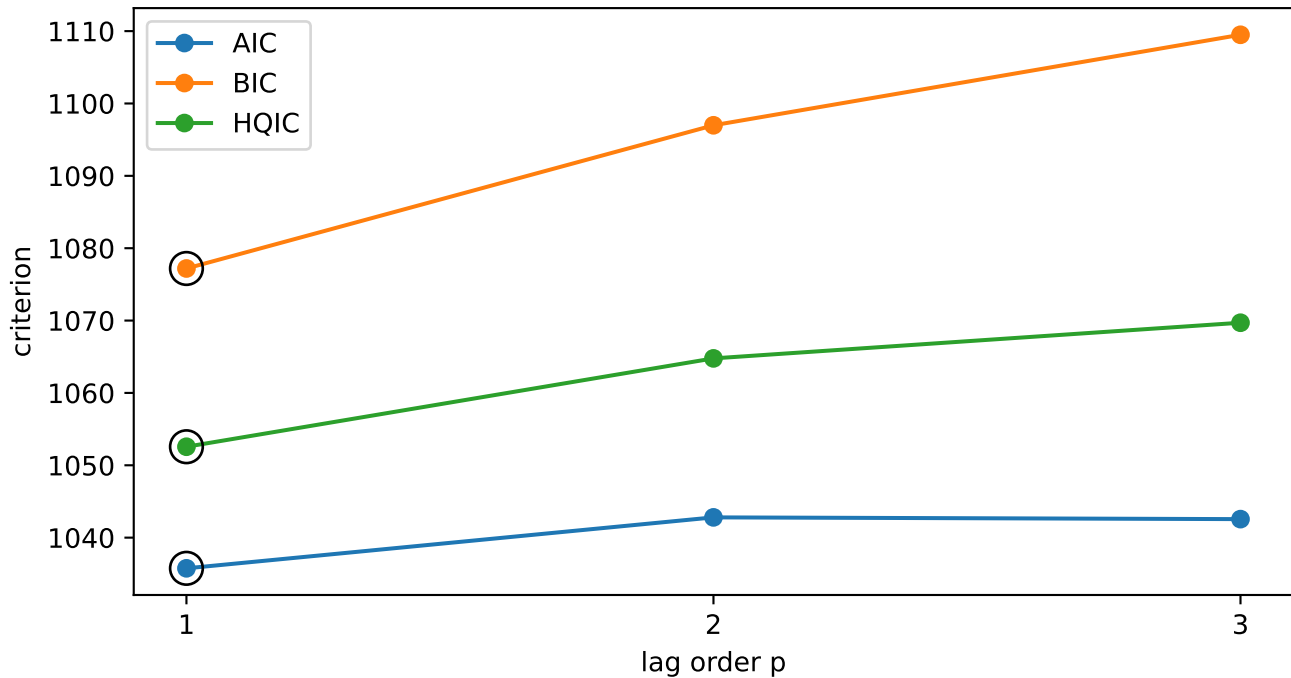
y2 (level) - ADF p=0.3336 [NON-stationary]

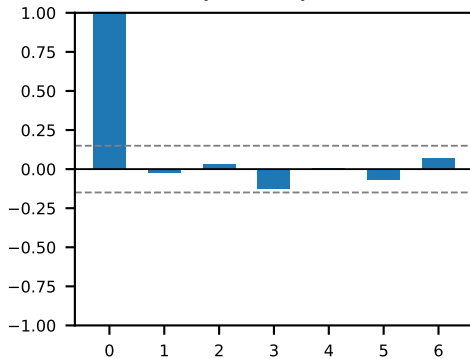
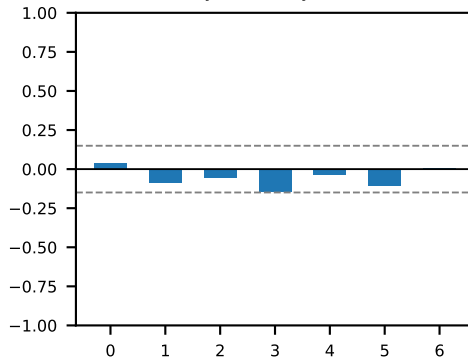
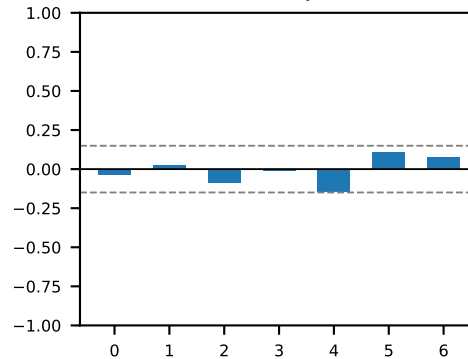
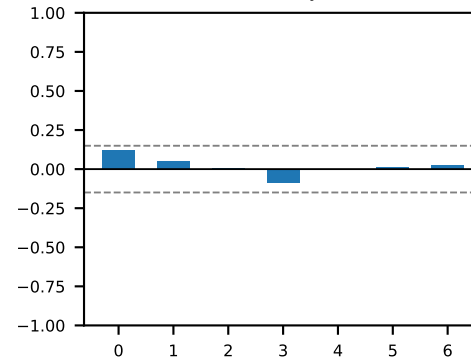
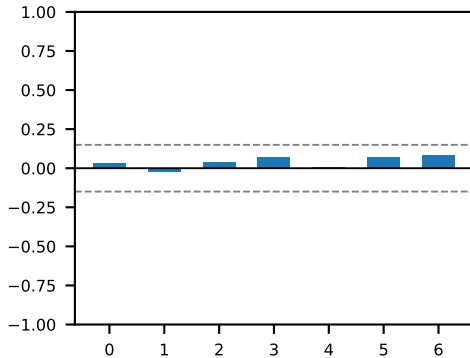
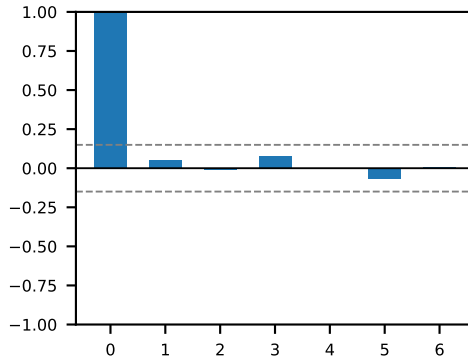
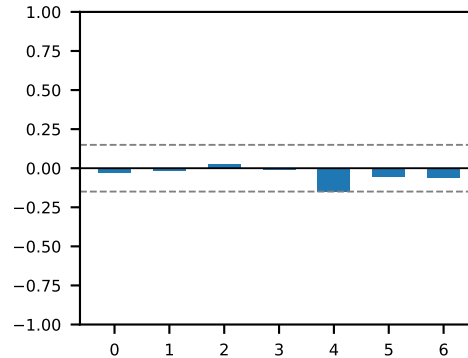


y2 (1st difference) - ADF p=0.0000 [stationary]



Lag order selection (circled = minimum)



$y1(t-k) \rightarrow y1(t)$  $y2(t-k) \rightarrow y1(t)$  $x1(t-k) \rightarrow y1(t)$  $x2(t-k) \rightarrow y1(t)$  $y1(t-k) \rightarrow y2(t)$  $y2(t-k) \rightarrow y2(t)$  $x1(t-k) \rightarrow y2(t)$  $x2(t-k) \rightarrow y2(t)$ 